EV charging system with RFID-based access control and billing mechanism.

01.	**ACS712 Current Sensor:** 1) **VCC:** Arduino 5V. 2) **GND:** Arduino GND. 3) **OUT:** Arduino **Analog Pin A0**. 4) **Terminal Blocks:** One side to **12V (+)**, the other side to the **Relay (COM pin)**.
02.	**Channel Relay Module:** 1) **VCC:** Arduino 5V. 2) **GND:** Arduino GND. 3) **IN1 / IN2:** Arduino **Digital Pins 8 and 7**. 4) **Output Side:** Connect the **Transmitter Coils** to the **NO (Normally Open)** terminals.
03.	**IR Sensors (Entrance, Slot 1, Slot 2):** 1) **VCC:** 5V | **GND:** GND. 2) **OUT (Entrance):** Arduino **Digital Pin 2**. 3) **OUT (Slot 1):** Arduino **Digital Pin 3**. 4) **OUT (Slot 2):** Arduino **Digital Pin 4**.

04.	**MFRC522 RFID Module:** 1) **VCC:** 3.3V (Do not connect to 5V). 2) **RST: Digital Pin 5.** 3) **GND: GND.** 4) **MISO: Digital Pin 12.** 5) **MOSI: Digital Pin 11.** 6) **SCK: Digital Pin 13.** 7) **SDA (SS): Digital Pin 10.**
05.	**I2C LCD Display:** 1) **GND: GND.** 2) **VCC: 5V.** 3) **SDA: Arduino Pin A4.** 4) **SCL: Arduino Pin A5.**
06.	**Servo Motor:** 1) **Brown Wire:** Ground (GND) 2) **Orange/Red Wire**: 5V 3) **Yellow/White Wire:** Digital Pin 9

07.	**Coil Connections (Relay to Coils):** 1) **Transmitter 1 (Slot 1):** Red wire – NO relay channel 1. 2) **Transmitter 2 (Slot 2):** Red wire – NO relay channel 2. 3) **Transmitter 1 (Slot 1):** Black wire – GND. 4) **Transmitter 2 (Slot 1):** Black wire – GND.
08.	**BUZZER:** 1) **Positive Terminal (+):** Digital Pin 6 2) **Negative Terminal (-):** GND (Ground)

- **12V Source:** Connect the positive and negative from the **DC Barrel Jack** to one set of power rails on the breadboard.
- **5V Logic:** Connect the **Arduino 5V** and **GND** pins to the other power rail.
- **Common Ground:** Connect the GND of the 12V rail to the GND of the 5V rail. This is vital for the sensors to work correctly.
- Connect the **Positive (+)** wire of your 9V/12V supply to the **COM (Common)** terminal of **both** Relay Channel 1 and Relay Channel 2. You can use a small jumper wire to "daisy-chain" the COM terminals together.